Synthetic lending repayment model

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Policy: to_tol0.010
Status: Academic experiment. No production release recommended.

Intended use

Demonstrate fairness assessment at scale for a Big Data case study. Intended users are the student, evaluator and technical reviewers. The model is unsuitable for live underwriting, pricing, credit limits or eligibility decisions.

Model and interface

Spark ML binary logistic regression uses income_z, debt_z, history_measured and employment_z. The inputs are synthetic standardized descriptors; they are not a production bureau schema. The positive label is would repay, and decision 1 means approve. The probability-like repayment score has not received a calibration audit.

Group membership is excluded from the logistic feature vector. Reweighing uses it at training time. Fairlearn postprocessing also uses it at decision time. Missing features, unknown groups and nonnumeric production data are outside the validated input range.

Data and training

2,000,000 synthetic applications; seed 327. Train/validation/test sizes are 1,200,876 / 399,890 / 399,234. The generator includes complete outcome labels, group-dependent feature distributions and a 0.9-unit under-measurement of group 1 credit history. It uses no Home Credit records or actual customers.

L2 logistic regularization: 0.01; maximum iterations: 40. Reweighing strengths: 0, 0.5, 1. ThresholdOptimizer uses 96,021 validation scores. 41 decision policies are compared. All policy selection uses validation data only.

Decision objective

Maximize validation accuracy with a TPR gap of at most 2 percentage points and approval between 20% and 90%. If infeasible, minimize the validation gap within those approval bounds. Fallback used: False. These are academic operating choices and have no claimed regulatory status.

Measured test performance

Metric	Baseline	Selected
Accuracy	77.49%	78.02%
Balanced accuracy	74.80%	75.55%
Equal opportunity gap	18.29 pp	0.73 pp
Equalized odds gap	30.22 pp	1.14 pp
Demographic parity gap	27.94 pp	6.33 pp
Approval rate	67.16%	66.44%
False positive rate	36.41%	34.77%
Default among approved	20.61%	19.89%

Selected subgroup	N	TPR	FPR	Approval
Group 0	219,954	86.16%	35.36%	69.28%
Group 1	179,280	85.43%	34.22%	62.96%

Equal opportunity measures approval differences among applicants who would repay. Equalized odds adds false-positive disparity. Both are reported as absolute gaps in percentage points; smaller is better. The default rate among approved applicants is a separate portfolio-risk measure.

Selected decision rule

The selected method is **threshold_optimizer**. Full settings are stored in results/policies.json and MODEL_CARD.md. Fairlearn threshold rules can mix two thresholds per group. The Spark implementation realizes the probabilities using a stable per-application ID hash, with seed salt 900. Calibration and held-out realized rates may differ.

Verification

Spark aggregates the complete test split. Fairlearn independently checks confusion-based group and overall metrics using weighted confusion cells. Maximum metric error: 0. Maximum threshold-export probability error: 0. These checks validate the implementation, not the representativeness of the synthetic population.

All figures refer to this fixed model, population and randomized-decision seed. They do not estimate uncertainty from model retraining or changes to the population.

Limitations and risks

This experiment has two fictional groups, one seed and one static synthetic population. It does not establish real-world accuracy, affordability or a reduction in actual discrimination. It does not evaluate intersectional groups, individual fairness, ranking quality, score calibration or temporal drift.

All synthetic applicants have labels. In real lending, repayment is usually observed only after approval and over a fixed outcome window. Rejected applicants must not automatically be assigned a default label. Missing or selectively observed outcomes can invalidate an apparently precise fairness estimate.

A lower TPR gap can coexist with unequal false positive rates or a larger default rate among approved applicants. This run has no loan amounts, pricing, recovery assumptions or quantified error costs. Its selected policy therefore is not a validated economic lending policy.

Group-dependent thresholds require protected-attribute data at decision time. Whether such use is appropriate needs review in the actual setting. Randomization also needs an accountable decision procedure, stable logging and an appeal process. A small metric gap is not evidence of legal compliance.

Conditions before any real pilot

Use representative data and a time-based holdout. Define repayment outcomes, maturity windows, loan costs and affordability constraints. Check calibration and errors across relevant groups and intersections. Repeat fitting across seeds and periods. Review protected-data handling and decision rules with appropriate domain reviewers.

Monitoring if a future model is deployed

Track approval rates, outcome maturity, subgroup sample sizes, TPR/FPR, default among approved, calibration and drift. Record model and policy versions, score inputs and randomized-decision seeds. Set alert thresholds and review ownership before rollout. Offer human review and a way to challenge adverse decisions.

Reproduction and ownership

Spark 3.5.3, Fairlearn 0.13.0, Python 3.11.16. The audit took 19.6 seconds on local[4] with 8 partitions. The package includes source, pinned dependencies, trained models, policy settings, count tables and the run manifest. Academic artifact owner: Kundhan.

Documentation structure: Mitchell et al. (2019), Model Cards for Model Reporting. https://arxiv.org/abs/1810.03993